Applied NLP

DATA 441/641

Lecture 12 — Deep dive into Chatbots

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Deep Dive into Components of a Dialog System

Understanding the context (i.e., the user response) in light of conversation history is one of the most important tasks for building a dialog system. Understanding context can be broken down into understanding the user's intent and detecting corresponding entities for that particular intent. These internal components correspond to the **natural language understanding** component in the chatbot pipeline.

User: I'm looking for a cheaper restaurant
`inform(price=cheap)`
System: Sure. What kind - and where?
User: Thai food, somewhere downtown
`inform(price=cheap, food=Thai, area=centre)`
System: The House serves cheap Thai food
User: Where is it?
`inform(price=cheap, food=Thai, area=centre); request(address)`
System: The House is at 106 Regent Street

Dialog Act Classification

Dialog act classification is a task to identify how the user utterance plays a role in the context of dialog. This informs what "act" the user is performing. For example, a simple example of dialog acts would be to identify a "yes/no" question.

Note: Building dialog act or intent classification from scratch can be a complex and data-consuming process. Doing so makes sense when our dialog acts and slots are more open-ended in nature than a Cloud API or existing framework can solve. Having complete control of dialog internals can yield better results over time in such problems.

Identifying Slots

Once we have extracted the intents, we want to move on to extracting entities. Extracting entities is also important for generating correct and appropriate responses to the user's input. We also saw in our Dialogflow example that extracting entities along with the intents creates a full understanding of the user's input.

In the example "I'm looking for a cheaper restaurant"—we want to identify "cheaper" as a price slot and take its value verbati —i.e., the value of the slot is "cheaper." (This is like entities extraction from sentences)

However, the disadvantages of such approaches is that they need a lot of labeled data for both intent and entity detection. Also, we need dedicated models for both of the tasks. This can make the system slow during deployment.

Response Generation

Once we identify the slots and intent, the final step is for a dialog system to generate an appropriate response. There are many ways to generate a response:

- **Fixed responses**: FAQ bots mainly use fixed responses. Based on the intent and values for the slots, a dictionary lookup is made on a pool of responses and retrieves the best response. A simple case would be to discard the slot information and have one response per intent. For more complex retrieval, a ranking mechanism can be built that ranks the pool of responses based on the detected intent and slot-value pairs (or the dialog state).

- **Use of templates**: To make responses dynamic, a templates-based approach is often taken. Templates are very useful when the follow-up response is a clarifying question.

- **Automatic generation**: More natural and fluent generation can be learned using a data-driven approach. Upon obtaining the dialog state, a conditional generative model can be built that takes a dialog state as an input and generates the next response for the agent. These models can be graphical models or DL-based language models.